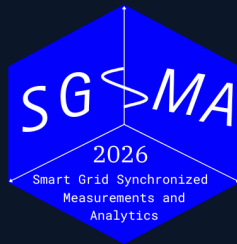


5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*



Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

Jorge L. Mayorga Taborda, Yessica Africano Rodriguez, Fernando Jimenez

Universidad de los Andes | SGSMA 2026

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Date

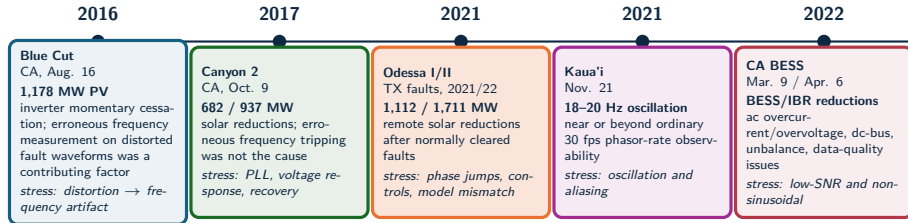
June 1-4, 2026

Venue

Santiago, Chile



Field events define the benchmark stress model



Benchmark implication: the estimator must be tested against waveform distortion, phase jumps, control recovery, unbalance, noise, harmonics, interharmonics, and reporting-rate limits, not only steady-state frequency error.

Source: NERC/WECC Blue Cut 2017; NERC/WECC Canyon 2 2018; NERC Odessa 2022; NERC/WECC CA BESS 2023; Dong et al. Kaua'i 2021 analysis

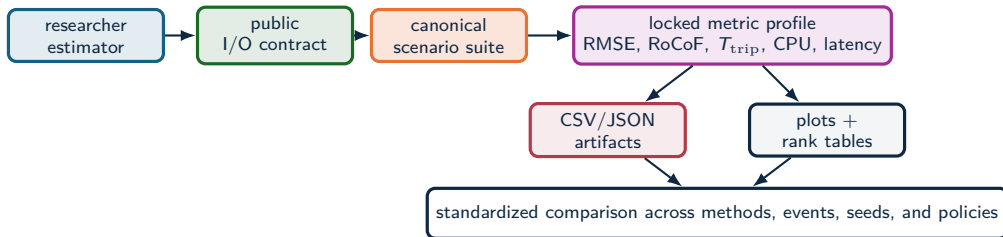
Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

OpenFreqBench platform



Researchers add

Estimator code, parameters, and hypotheses through the public contract.

Principle: a submitted estimator becomes a repeatable benchmark entry under the same scenarios, seeds, metrics, and artifacts.

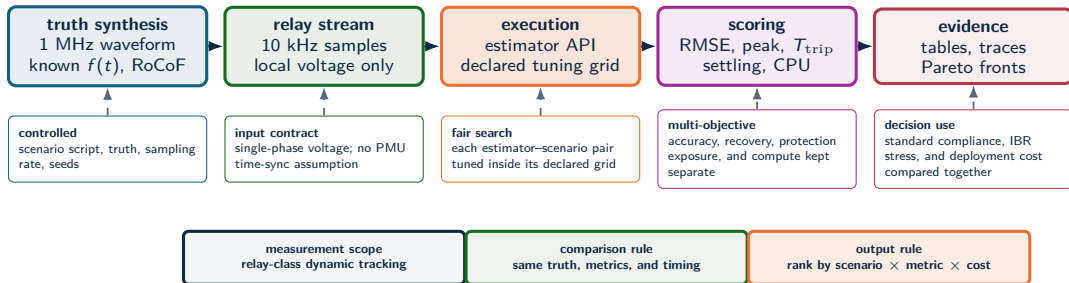
Platform fixes

Truth signals, sampling, seeds, tuning policy, metric formulas, and schema.

Outputs

Benchmark reports, metric tables, plots, manifests, and readiness gates.

Experimental method

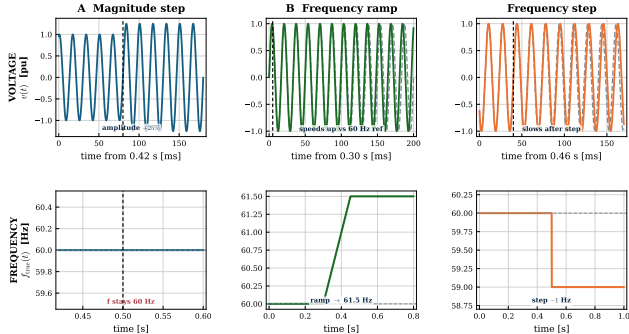


Source: Methodology and test framework

Standard stresses (1 of 2): magnitude, ramp, frequency step

CORE SCENARIO SUITE

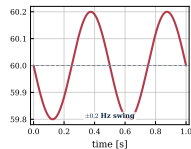
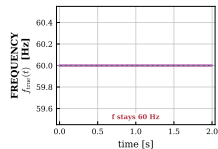
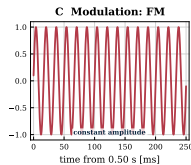
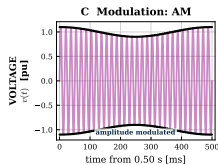
Standard stresses (1 of 2): VOLTAGE $v(t)$ on top, true FREQUENCY $f(t)$ below



Standard stresses (2 of 2): modulation (AM and FM)

CORE SCENARIO SUITE

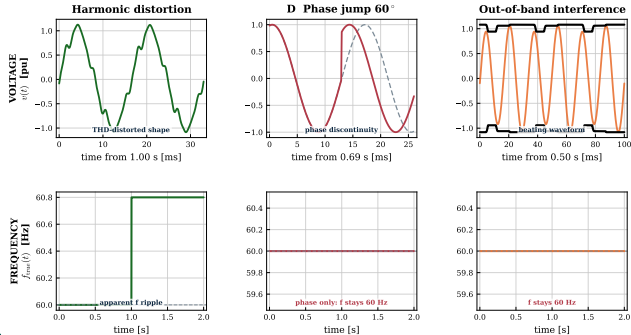
Standard stresses (2 of 2): VOLTAGE $v(t)$ on top, true FREQUENCY $f(t)$ below



Composite IBR stresses (1 of 2): harmonics, phase jump, out-of-band

IBR STRESS SUITE

Composite IBR stresses (1 of 2): VOLTAGE $v(t)$ on top, true FREQUENCY $f(t)$ below

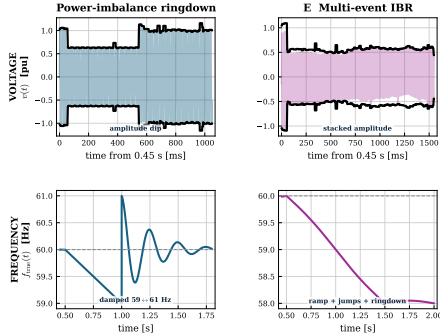


Top row voltage, bottom row true frequency: harmonics, a phase jump and

Composite IBR stresses (2 of 2): ringdown and multi-event

IBR STRESS SUITE

Composite IBR stresses (2 of 2): VOLTAGE $v(t)$ on top, true FREQUENCY $f(t)$ below



Scenario families: distortion, leakage, and low SNR

Compact signal model

$$v(t) = A(t) \sin \theta(t) + v_h(t) + v_x(t) + n(t)$$

$$v_h(t) = \sum_{h=2}^H a_h \sin(h\theta + \phi_h), \quad v_x(t) = \sum_q b_q \sin(2\pi f_q t + \psi_q)$$

$$\text{THD} = \frac{\sqrt{\sum_{h \geq 2} V_h^2}}{V_1}, \quad \text{SNR}_{dB} = 20 \log_{10} \left(\frac{1/\sqrt{2}}{\sigma_n} \right)$$

Interpretation

The benchmark keeps the truth signal and the waveform perturbation separate. When $f_{\text{true}}(t)$ is fixed, estimator error measures artifact rejection, not a physical frequency event.

Family	What it tests
Harmonics	Integer multiples of 60 Hz; THD rejection with fixed true frequency.
Interharmonics	Non-integer tones; leakage and beating that can mimic dynamics.
OOB tones	Out-of-band interference; filter selectivity, aliasing margins, PLL leakage.
Noise / SNR	Controlled σ_n ; low-SNR crossings and covariance assumptions.

Why this slide matters

In these cases $f_{\text{true}}(t)$ can remain nominal. Any frequency excursion reported by the estimator is then a waveform-induced measurement artifact.

Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL,
Type-3 SOGI-PLL.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF,
LKF2.

Legacy baselines

TKEO and ZCD are retained as
comparison references outside the
recommended deployment set.

Window / spectral

IpDFT, TFT, ESPRIT, Prony,
MUSIC.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Interpretation rule

These families occupy different
latency and stress-sensitivity
positions. The comparison tracks
where each family wins, degrades,
and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

Metrics and decision variables

Error and recovery

$$e[k] = \hat{f}[k] - f_{\text{true}}[k]$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_k e[k]^2}$$

$$E_{\text{max}} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\text{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5 \text{ Hz}\}$$

$$t_s = \min\{t : |e[\tau]| \leq \epsilon, \forall \tau \geq t\}$$

$$C_{\text{CPU}} = \text{mean time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	0.006	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	0.028

RMSE [Hz], mean over 5 seeds. Green:

below 0.05 Hz guide; red: failure/divergence regime.

What the heatmap shows

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

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UKF	0.000	0.017	0.055
LKF2	0.138	0.015	0.028
PLL	0.033	1.47	0.227
IpDFT	0.000	14.4	0.114
EKF	0.000	37.1	0.064

RMSE [Hz], mean over 5 seeds. Green:

below 0.05 Hz guide; red: failure/divergence regime.

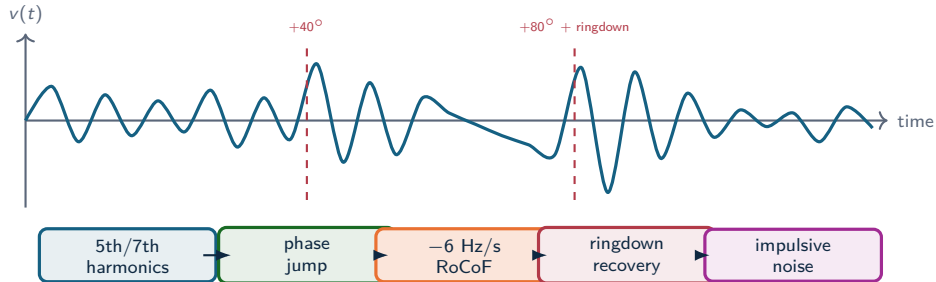
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Composite IBR sequence: why it is hardest



Why standard tests miss it

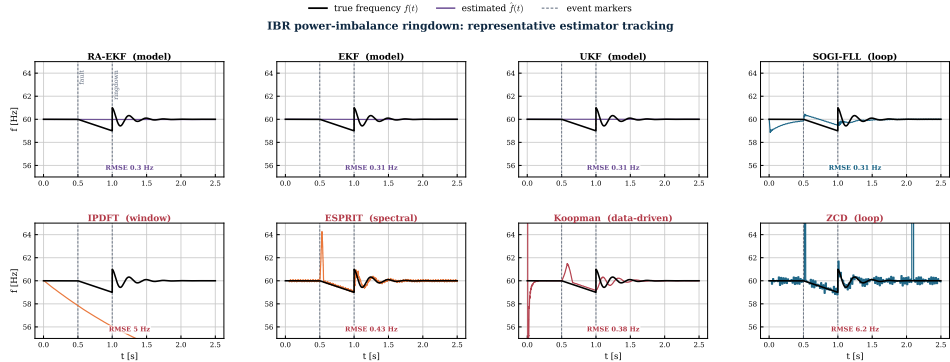
The estimator sees spectral leakage, loop re-lock, state-model mismatch, transient recovery, and noise in one nonstationary waveform.

Why ATLAS decomposes it

The multi-event case proves the failure exists; severity sweeps identify which stressor actually breaks each estimator family.

How estimators track an IBR ringdown

FULL MONTE-CARLO SIGNALS



Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step ≈ 0 ; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF is the most robust ramp tracker	0.006 Hz vs EKF 37 Hz, IpDFT 14 Hz (diverge)	Divergence is intermittent (seed-dependent)
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\text{trip}} = 0$; IpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all ≈ 1.1 Hz	UKF 1.09 Hz best; ZCD 426 Hz diverges	RMSE hides trip-risk and 1000 $\times$ CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 $\mu\text{s}/\text{sample}$; Koopman $\sim 15,000$	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Computational feasibility

Method	Time/sample	Family	Target
SOGI-FLL	0.0348 μ s	Loop	MCU
SOGI-PLL	0.0867 μ s	Loop	MCU
EKF	0.813 μ s	Model	MCU/DSP
RA-EKF	1.61 μs	Model	DSP
TFT / IpDFT	3–4 μ s	Window	DSP/FPGA
Koopman	390 μ s	Data-driven	CPU/GPU
ESPRIT	2,838 μ s	Window	CPU

Source: SGSMA benchmark matrix, Sec. IV-D and Fig. 2(g)-(h)

Deployment interpretation

CPU is a software-cost indicator from the benchmark harness, not a hardware-latency claim.

Key comparison

The practical low-cost cluster is SOGI-FLL, SOGI-PLL, EKF, and RA-EKF. Koopman and ESPRIT sit far to the right on cost without a global RMSE gain.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P		
TFT	P	P	P		
IpDFT	P	F	P		
SOGI-FLL	P	P	P		
PLL	P	F	F*		
EKF	P	F	P		
UKF	P	P	P		
RA-EKF	P	P	P		
LKF2	F*	P	P		

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/ T_{trip} /settling threshold | **F*** borderline.

Compliance heatmap: why standard tests are insufficient

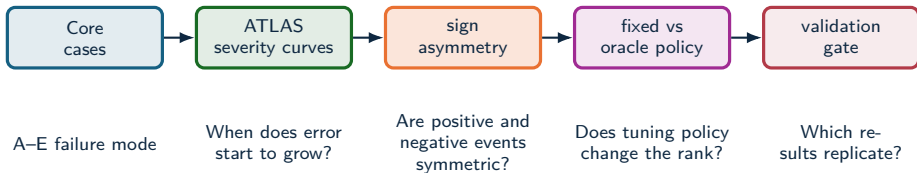
Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P	F	F
TFT	P	P	P	F*	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	P	P	P	F
PLL	P	F	F*	P	F
EKF	P	F	P	P	F
UKF	P	P	P	P	F
RA-EKF	P	P	P	P	F
LKF2	F*	P	P	P	F

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/ T_{trip} /settling threshold | **F*** borderline.

Result: Step, ramp, and modulation compliance does not predict phase-jump or multi-event readiness.

ATLAS severity analysis

ATLAS PAPER-GRADE SWEEP: $n = 30$, fixed policy, 9/9 sweeps — 17 canonical estimators + MUSIC extra



ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: docs/ATLAS_PIPELINE.md; docs/ATLAS_METHOD_AUDIT.md

ATLAS result: where estimators stop being valid

ATLAS PAPER-GRADE SWEEP: $n = 30$, fixed policy, 9/9 sweeps — 17 canonical estimators + MUSIC extra

Stress family	Maximum tested level	Methods below 0.05 Hz	Interpretation
Magnitude step	$\pm 90\%$ sag/swell	2	EKF and RA-EKF survive both signs.
AM modulation	10 Hz envelope	10	AM alone is not the hardest modulation case.
FM modulation	10 Hz modulation	0	FM is qualitatively harder than AM.
RoCoF ramp	± 50 Hz/s	0	Severe RoCoF requires a separate qualification class.
Phase jump	$\pm 60^\circ$	0	Single jump severity already breaks the 0.05 Hz guide.
Harmonics	30% THD	0	Low-THD winners do not survive extreme THD.
Interharmonics	20% at 75 Hz	0	Integer-harmonic tests are not sufficient.
Broadband noise	$\sigma = 0.1$ pu	0	Noise robustness is a first-class stress dimension.

ATLAS stability atlas: fast categorical read

ATLAS PAPER-GRADE SWEEP: $n = 30$, fixed policy, 9/9 sweeps — 17 canonical estimators + MUSIC extra

ATLAS stability atlas: categorical operating-limit view

estimator	Free step	RoCoF	Phase jump	Mag step	Harmonics	Inter-harm.	Noise	AM	FM
PLL	fragile 0.01	fragile 0.10	fragile 3	fragile 2	fragile	fragile 0.30	fragile 1e-34	fragile 0.10	fragile 0.10
SOGI-PLL	fragile 0.50	fragile 10	fragile	fragile	fragile	fragile 0.50	fragile 0.01	usable 10	usable 0
SOGI-FLL	fragile 0.50	fragile	fragile	usable 3	fragile	fragile	fragile	usable 10+	fragile
T3-SOGI-PLL	fragile 0.01	fragile 0.10	fragile 3	fragile 2	fragile	fragile 0.30	fragile 1e-34	fragile 0.10	fragile 0.10
ZCD	fragile 0.50	fragile	fragile	usable 00	usable 20	fragile	fragile 1e-34	usable 10+	usable
IPDFT	fragile 0.50	fragile 3	fragile 3	fragile 35	fragile 3	fragile	fragile 0.01	stable 10+	usable 5
TFT	fragile 0.50	fragile	fragile	usable 00	fragile	fragile	fragile 0.01	stable 10+	usable
ESPRIT	fragile 3	fragile 10	fragile 3	fragile 20	fragile 2	fragile 0.30	fragile 0.01	stable 10+	usable 0
Prony	invalid flag	invalid flag	invalid flag	invalid flag	fragile	fragile 0.30	invalid flag	fragile 0.10	fragile 0.10
MUSIC	fragile	fragile 10	fragile	fragile	fragile	fragile 0.30	fragile 0.01	stable 10+	usable 0
EKF	fragile 0.30	fragile 0.30	usable 10	usable 00+	usable 15	fragile	usable 0.10	stable 10+	fragile 0.10
UKF	fragile 0.50	fragile 10	fragile	usable	usable	fragile	usable 10+	stable 10+	fragile
RA-EKF	fragile 0.50	fragile 3	fragile 10	stable 00+	fragile	fragile 3	usable 0.10	stable 10+	fragile 2
LKF	fragile 0.50	fragile 0.10	fragile	fragile	fragile	fragile 0.30	fragile 1e-34	fragile 0.10	fragile 0.10
LKF2	fragile 0.50	fragile 0.10	fragile 10	usable 50	fragile 2	fragile 2	fragile 0.01	stable 10+	fragile 0.10
RLS	fragile 0.01	fragile 0.10	invalid flag	invalid flag	invalid flag	invalid flag	invalid flag	fragile 0.10	fragile 0.10
TKEO	fragile 0.01	fragile 0.10	fragile 3	fragile 2	fragile 2	fragile 2	fragile 1e-34	fragile 0.10	fragile 0.10
Koopman	fragile 0	fragile 10	fragile	fragile 0	fragile	fragile	fragile 0.01	stable 10+	usable 2

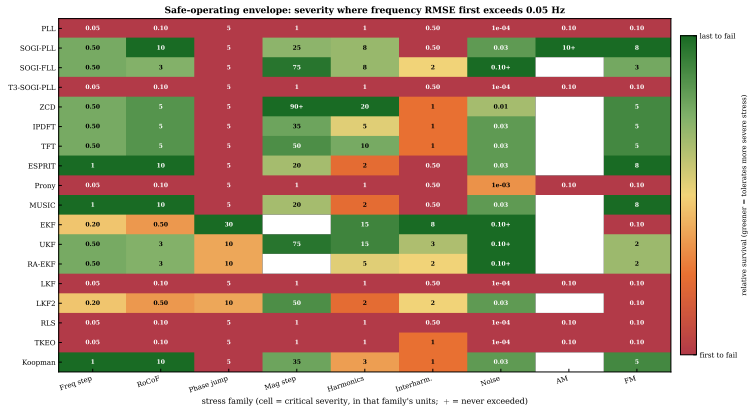
stress family

stable RMSE <= 0.05 Hz across tested range
usable threshold reached after >=50% of max severity
fragile threshold crossed early
invalid/diverged non-finite, invalid-rate, or bound-hit flag

Cell value is the first severity where frequency RMSE exceeds 0.05 Hz, in each stress family's units; '+' means the guide was not exceeded. ATLAS artifact: 17 canonical estimators plus MUSIC extra; PI-GRU is handled separately.

Safe-operating envelope

ATLAS PAPER-GRADE SWEEP: $n = 30$, fixed policy, 9/9 sweeps — 17 canonical estimators + MUSIC extra



AM, FM, and interharmonics split the story

ATLAS PAPER-GRADE SWEEP: $n = 30$, fixed policy, 9/9 sweeps — 17 canonical estimators + MUSIC extra

Stress	Winner pattern	Threshold behavior
AM modulation	EKF wins slow, reference, fast, and severe regions.	Ten methods stay below 0.05 Hz through 10 Hz envelope modulation.
FM modulation	SOGI-FLL, TFT, and Koopman split the low-rate points; ESPRIT leads from 1–10 Hz.	No method stays below 0.05 Hz at 10 Hz FM modulation.
Interharmonics	EKF leads 0.5–15%; PLL leads at 20%.	EKF crosses the 0.05 Hz guide between 5% and 8%; most families cross earlier.

Implication

AM, FM, integer harmonics, and interharmonics should not be collapsed into one “distortion” label. They select different estimator families.

Decomposition insight

A composite IBR trace can contain AM, FM, harmonics, interharmonics, noise, and phase steps. ATLAS isolates which stressor changes the ranking.

When the textbook filter diverges

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

Estimator	seed 1	seed 2	seed 3	seed 4	seed 5	trip-risk
EKF (textbook)	0.016	63.7	0.017	121.5	0.027	up to 1.35 s
RA-EKF	0.0055	0.0062	0.0083	0.0045	0.0058	0 s

What happens

The standard EKF tracks most seeds but **intermittently diverges** to tens of Hz. It is *not* a severity threshold: the worst seed (121 Hz) had a low ramp rate (1.7 Hz/s) while the steepest (5 Hz/s) tracked cleanly. The failure is triggered by onset/phase combinations — seed-dependent and silent.

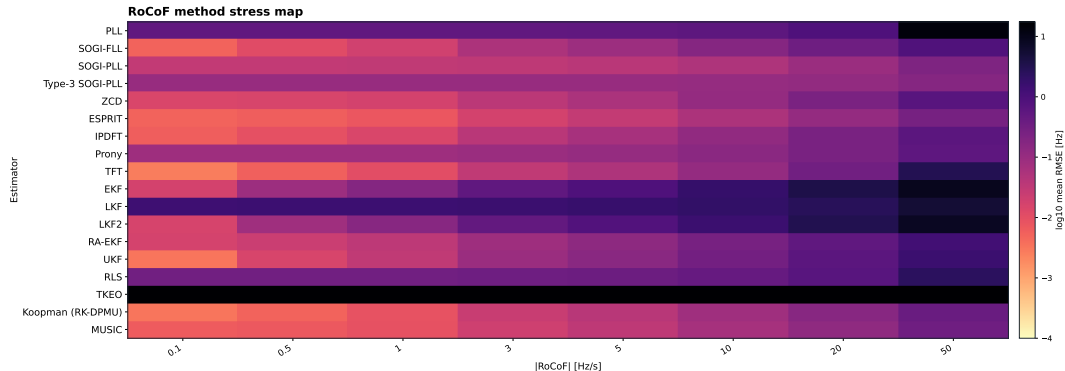
Why RA-EKF matters here

Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.

No estimator owns the whole stress space

ATLAS PAPER-GRADE SWEEP: $n = 30$, fixed policy, 9/9 sweeps — 17 canonical estimators + MUSIC extra

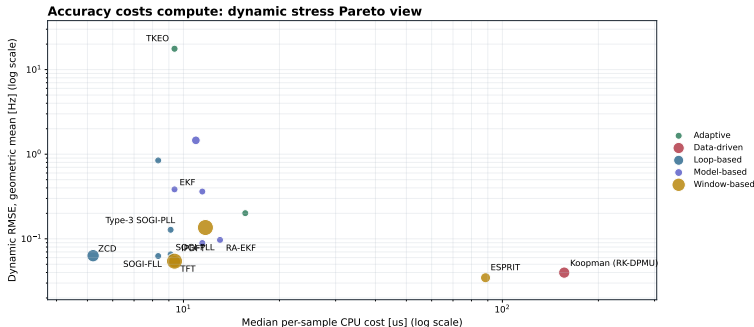
RoCoF winners move by regime: Koopman at low stress, ESPRIT through the standard/IBR band, SOGI-PLL at severe RoCoF.



Accuracy costs 100–1000× more compute

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario

Deployable methods sit lower-left; spectral/data-driven accuracy moves right by 100–1000× CPU.



Three findings

1. Metric-conditioned

The SGSMA matrix and ATLAS both show that RMSE, trip exposure, RoCoF, and CPU can select different methods.

2. Stress-bounded

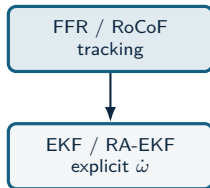
AM remains survivable for many methods; severe FM, RoCoF, interharmonics, noise, and extreme THD break the 0.05 Hz guide.

3. Robustness over point accuracy

The OpenFreqBench ramp seeds expose silent divergence; RA-EKF removes that failure mode in the tested run.

Dynamic frequency estimators should be qualified by event family, sign, severity, recovery, trip exposure, and compute envelope.

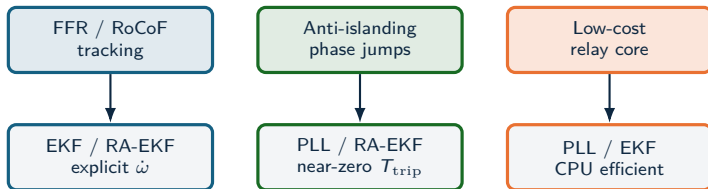
Recommendation map



Recommendation map



Recommendation map



Recommendation map



Recommendation map



Deployability depends on event class, trip-risk tolerance, and compute budget.

Validation agenda and conclusion

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

Result: estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.